J. KENISHA JUDITH

192572071

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	Value
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

4) streamWave - Iaas, Paas, Saas and Xaas

Introduction:

Developments in hardware, virtualization and internet technologies have enabled organizations to access computing resources and software through cloud computing.

1. Developments in computing Hardware

* High-performance processors improve processing speed.
* Increased RAM supports multiple virtual machines.
* SSDs provide faster storage.
* High-speed networks improve communication.
* Virtualization improves hardware utilization.

2. Developments in Internet software

* Web browsers provide access to cloud applications.
* REST APIs enable application communication.
* Web services support integration.
* Automation tools simplify deployment and management.

3. cloud service models :

A. IaaS (Infrastructure as a Service)

Provides virtual machines, storage and networking

★ users control operating systems and installed software.

★ suitable for StreamWave's existing content-management application.

Eg:- Amazon EC2, Azure virtual machine.

B. Paas (Platform as a service)

★ Provide a development and deployment platform.

★ No need to maintain servers or runtime environment.

★ supports application development and automatic scaling.

★ suitable for StreamWave's new customer-facing application.

eg:- Google App Engine, AZure App Service

C. Saas (Software as a service)

Provides ready-made software through the internet.

★ Accessible through a browser.

2) given:

* old servers = 15
* CPU utilization = 25%
* old server power = 350 W
* New host power = 450 W
* Target utilization = 65%
* N+1 = 1 host
* Electricity cost = ₹ 1/ kWh
* operating hours = 8760/year

(a) Active hosts and total hosts

Total workload:

$$15 \times 0.25 = 3.75$$

Active hosts:

$$\frac{3.75}{0.65} = 5.77 \approx 6$$

Including N+1:

$$6+1 = 7 \text{ hosts}$$

(b) consolidation ratio:

$$\frac{15}{6} = 2.5 : 1$$

(c) CPU utilization

All 7 hosts active:

$$\frac{3.75}{7} \times 100 = 53.57\%$$

After one host fails:

$$\frac{3.75}{6} \times 100 = 62.5\%.$$

(d) Annual electricity saving

old power:

$$15 \times 350 = 5250\,W = 5.25\,KW$$

New power:

$$7 \times 450 = 3150\,W = 3.15\,KW$$

old annual cost:

$$5.25 \times 8760 \times 9 = ₹\,4,13,910$$

New annual cost:

$$3.15 \times 8760 \times 9 = ₹\,2,48,346$$

Annual saving:

$$413910 - 248346$$

$$₹\,1,65,564$$

Assumptions: Equal workload on all odd servers, all seven new hosts are powered on, and cooling costs are excluded.

Conclusion: Server consolidation reduces electricity consumption while N+1 provides fault tolerance.

given :

* Physical CPU cores = 24
* RAM = 96 GiB
* Hypervisor reservation = 8 GiB
* VMs = 12
* Each VM = 3 vCPU and 8 GiB RAM
* Normal RAM usage = 5 GiB per VM
* Normal CPU usage = 35%
* Peak CPU usage = 75%

(a) over-commitment ratios

Total vCPU :

$$12 \times 3 = 36$$

vCPU ratio :

$$\frac{36}{24} = 1.5 : 1$$

Usable RAM :

$$96 - 8 = 88 \text{ GiB}$$

Allocated RAM :

$$12 \times 8 = 96 \text{ GiB}$$

Memory ratio :

$$\frac{96}{88} = 1.09 : 1$$

(b) Memory sufficient

Normal memory usage :

$$12 \times 5 = 60 \text{ GiB}$$

Available RAM = 88 GiB

∴ The memory is sufficient

Peak memory usage

$$(8 \times 8) + (4 \times 6)$$
$$= 64 + 24 = 88\,GB$$

∴ peak memory is exactly equal to available RAM, with no free memory

(c) CPU sufficiency

Normal CPU usage : $36 \times 0.35 = 12.6$

$$\frac{12.6}{24} \times 100 = 52.5\%$$

CPU is sufficient

Peak CPU usage:

$$36 \times 0.75 = 27$$

$$\frac{27}{24} \times 100 = 112.5\%$$

CPU demand exceeds available capacity by 3 core - equivalents. Therefore CPU is sufficient during peak operation.

(d) Performance and security issues with remedies

performance issues :

* CPU contention
* memory shortage and swapping
* Increased application response time

security issues :

* Unauthorized VM access
* Data Leakage

* Software - defined manages research data.
* software - defined networking controls communication.
* Automation manages workloads.

It improves scalability, flexibility and resource utilization.

3. Role of SOA :-
SOA enables independent applications to communicate through standard interfaces
It supports sharing of :-
* Research - computing services
* Digital - library services
* Authentication services
* Data access services.

REST APIs and SOAP enable communication between universities.

4. challenges and solution :-

challenges	solution
Resource sharing	standard APIs and protocols
service availability	Redundant servers and load balancing

Remedies :

* Add CPU cores and RAM.
* Monitor resource utilization.
* Apply resource limits.
* use firewalls and encryption.
* Implement MFA and access control.

4) cloud federation connects independent cloud data centers to share computing resources and services. Two universities can share research-computing and digital library resources through federation.

1. conventional cloud data center

* Servers host research application.
* storage maintains research datasets and digital books.
* Networking connects both universities.
* virtualization runs multiple applications.
* Management software ~~manage~~ monitors resources.

2. software defined- Data center (SDDC)

SDDS manages resources through software

* software-defined compute allocates processing resources.

Federated Identity — SAML, openID connect and SSD

User presence privacy — consent and restricted areas

5. Technical and Administrative safeguards

Technical:
* MFA and RBAC
* Data encryption
* Firewalls and network segmentation
* security monitoring
* Backup and disaster recovery

Administrative:
* Federation agreements
* Data sharing policies
* Regular security adults
* Employee training

5) given:-
* cameras = 150
* clip size = 4MB
* one clip every 3 minutes
* storage cost = ₹ 1.50/GB/month
* serverless cost = ₹ 0.00015/invocation
* Filtering = 25%

* 1GB = 1000 MB

(a) Daily data volume and bandwidth:

clips per camera per day:

$$\frac{24 \times 60}{3} = 480$$

Total clips: $150 \times 480 = 72000$

Daily data: $72000 \times 4 = 288000$ MB

288 GB/ day

Average bandwidth: $\frac{288000 \times 8}{86400} = 26.67$ Mbps

(b) Monthly storage cost without filtering

$288 \times 30 = 8640$ GB

$8640 \times 1.50 =$ **₹ 12,960**

(c) After edge filtering

Daily data: $288 \times 0.25 = 72$ GB

Bandwidth: $26.67 \times 0.25 = 6.67$ Mbps

Monthly storage: $72 \times 30 = 2160$ GB

storage cost: $2160 \times 1.50 =$ ₹ 3,240

Monthly saving: $12960 - 3240 =$ ₹ 1,720

(d) Monthly serverless invocation cost

monthly invocations: $72000 \times 30 \times 0.25 = 540000$

cost: 540000×0.00015

₹81